DWG. No. 310SC90-775

THIS DOCUMENT IS THE PROPERTY OF MITSUBISHI HITACHI POWER SYSTEMS, LTD. (MHPS) AND CONFIDENTIAL. IT MUST NEITHER BE COPIED NOR USED FOR ANY PURPOSE OR MANNER OTHER THAN EXPRESSLY PERMITTED BY MHPS OR WITH A PRIOR WRITTEN CONSENT OF MHPS NEITHER IS IT TO BE HANDED OVER NOR COMMUNICATED TO A THIRD PARTY IN ANY WAY. BREACH OR INFRINGEMENT OF ANY KIND WILL LEAD TO LEGAL ACTION.

REVISIONS

REV.	DESCRIPTION	REVD.	CHKD.	APPD.	DATE
B	TO REFLECT THE DESIGN PROGRESS AND CUSTOMER COMMENTS	R. KOJIMA	M. SAKAKIBARA	N. MATSUMOTO	2013-12-06
C	TO REFLECT THE DESIGN PROGRESS AND CUSTOMER COMMENTS	R. KOJIMA	M. SAKAKIBARA	N. MATSUMOTO	2014-02-13
D	TO REFLECT THE DESIGN PROGRESS	R. KOJIMA	M. SAKAKIBARA	N. MATSUMOTO	2014-10-02
E	TO REFLECT THE DESIGN PROGRESS	M. SAKAKIBARA	N. MATSUMOTO	N. MATSUMOTO	2014-11-14
0	SUBMISSION AS "FINAL" ENGINEERING DOCUMENT	R. KOJIMA	M. SAKAKIBARA	M. SAKAKIBARA	2015-05-29

ATS UNIT SEQUENCE CONTROL PROGRAM

三菱日立パワーシステムズ株式会社
2015-05-29
MITSUBISHI HITACHI POWER SYSTEMS, LTD.

FOR FINAL

REFERENCE DRAWINGS

1. ATS CRITERIA LIST AND CRITERIA TABLE (310SC90-776)
2. TURBINE GENERATOR INTERLOCK BLOCK DIAGRAM (310SC90-774)
3. GENERATOR PROTECTION/BREAKER LOGIC DIAGRAM (310SD03-743)
4. START-UP AND SHUTDOWN CURVES (IP TURBINE START-UP) (PPE-13-0351)

CONFIDENTIAL

FORMOSA HEAVY INDUSTRIES CORP.
100, SWEIKUAN RD., JENWU, KAOHSIUNG, TAIWAN, R.O.C.

PROJECT TITLE / LOCATION

SMC DAVAO POWER PLANT PROJECT
MALITA, PROVINCE OF DAVAO DEL SUR, PHILIPPINES

OWNER / ADDRESS

San Miguel Consolidated Power Corporation
40 SAN MIGUEL AVENUE, MANDALUYONG CITY
1550 METRO MANILA, PHILIPPINES

Drawn : Name/Date	R. KOJIMA	/2013-06-06
Checked : Name/Date	M. SAKAKIBARA	/2013-06-06
Approved: Name/Date	S. NEMOTO	/2013-06-06

Title,supplementary title: ATS UNIT SEQUENCE CONTROL PROGRAM

Responsible dep.: Hitachi I&C Engineering Section/ Instrument & Control Department

Scale: 1 : 1

DWG. No.	Rev.	Sheet	Date of issue
310SC90-775	0	1/31	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

CAD SOFTWARE: AutoCAD
SIZE DESIGNATION: A3

PAGE NO.	DESCRIPTION	PAGE NO.	DESCRIPTION	PAGE NO.	DESCRIPTION
SH.1	COVER	19~22	ACCELERATION		
1	CONTENTS	23	EXCITATION		
2	GENERAL, SYMBOL & ABBREVIATION	24~25	SYNCH. & INITIAL LOAD		
3	DESCRIPTION OF ATS PROGRAM	26~27	LOAD UP		
4	EXPLANATION FORMAT	28	NORMAL OPERATION		
5	CONTROL STEP TRANSITION & STATE TRANSITION	29	TURBINE SHUT DOWN		
6	ATS AUTO MODE/OFF MODE LOGIC	30	HOUSE LOAD OPERATION INDICATION		
7~8	SEQUENCE PROGRESS LOGIC				
9	CRITERIA CHECK LOGIC				
10	CRITERIA BYPASS LOGIC				
11	SPEED/LOAD HOLD LOGIC				
12	START UP MODE SELECT				
13	HP TURBINE WARMING				
14~16	TURBINE PREPARATION				
17~18	TURBINE RUB CHECK				

TYPE –					PAGE 1	
SERVICE CONTENTS						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1 : 1		DWG. No. 310SC90-775	Rev. 0	Sheet 2	Date of issue 2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

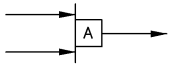
GENERAL

This drawing shows the outline of control and the conditions for lighting or not lighting of the lamps displayed on the TCS HMI.

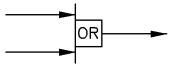
SYMBOL

ELECTRICAL SIGNAL

COMMUNICATION LINK

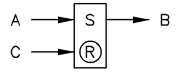


"AND" GATE

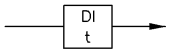


"OR" GATE

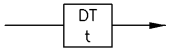
"NOT" GATE



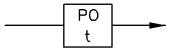
MEMORY CIRCUIT
IF THERE IS NOT "C" SIGNAL,
"A" SIGNAL IS SELF-HOLDED
AND IT GOES TO "B".
IF THERE IS "C" SIGNAL,
THE SELF-HOLD IS RESET AND
"A" SIGNAL DOES NOT GO TO "B".
("C" SIGNAL HAS PRIORITY)



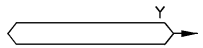
TIMER (DELAY PICK UP)
t: SETTING TIME



TIMER (DELAY DROP OUT)
t: SETTING TIME



TIMER (ONE SHOT)
t: SETTING TIME



CONDITION
Y : INSTRUMENT TAG No.



LOGIC CONTINUATION (DESTINATION)
A : CONNECTION POINT
N : PAGE No.



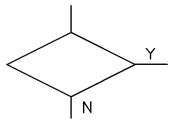
LOGIC CONTINUATION (DEPARTURE)
A : CONNECTION POINT
N : PAGE No.


PROGRAM START


PROGRAM STOP

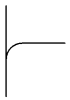

LAMP ON


LAMP OFF



QUESTION


AND LOGIC


OR LOGIC


CONTROL IN ATS


OUT OF MHPS'S SCOPE



TIME DELAY

ABBREVIATION

THSC	:	THERMAL STRESS CALCULATION
SYNCH.	:	SYNCHRONIZING
CV	:	CONTROL VALVE
MSV	:	MAIN STOP VALVE
TEMP.	:	TEMPERATURE
DIFF.	:	DIFFERENTIAL
CRI.	:	CRITERIA OR "ATS CRITERIA LIST AND CRITERIA TABLE" (OUR DWG. NO.310SC90-776)
INIT	:	INITIAL
HP	:	HIGH PRESSURE
LP	:	LOW PRESSURE
RH	:	REHEAT
Y	:	YES
N	:	NO
AVR	:	AUTOMATIC VOLTAGE REGULATOR
PB	:	PUSH BUTTON
ASD	:	AUTOMATIC SYNCHRONIZING DEVICE
GEN.	:	GENERATOR
G.C.B.	:	GENERATOR CIRCUIT BREAKER
T-G	:	TURBINE GENERATOR
AUX.	:	AUXILIARY
SEQ.	:	SEQUENCE
TRANSF.	:	TRANSFER
MIN.	:	MINIMUM
CCS	:	CO-ORDINATED CONTROL SYSTEM
M.T.	:	MORE THAN
L.T.	:	LESS THAN
UCP	:	UNIT CONTROL PANEL
TCS	:	TURBINE CONTROL SYSTEM
ATS	:	AUTOMATIC TURBINE START-UP/SHUT-DOWN
DCS	:	DISTRIBUTED CONTROL SYSTEM
GCP	:	GENERATOR CONTROL AND METERING PANEL

TYPE	–				PAGE	2
SERVICE	GENERAL, SYMBOL & ABBREVIATION					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-775	0	3	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

ATS PROGRAM

Automatic Turbine Start-up/ Shut-down System (ATS) is a sequential control program to assist an operator in operating the turbine start-up/shut-down.

During the sequential control process, ATS sends a pulse signal to "SUBLOOP" or "AUX. EQUIPMENT" by communication link as necessary. ATS also use the hardwired cable for a continuous signal depending on the sequential control requirement.

The detail functions of ATS are described as follows:

1. ATS AUTO MODE / OFF MODE

(1) When an operator selects "ATS AUTO MODE" PB from TCS/DCS HMI, ATS program starts to monitor the criteria (stated in "4. CRITERIA CHECK") and starts to send the output command signals to each step control program for automatic sequential operation.
(2) When the operator selects "OFF MODE" PB from TCS/DCS HMI, ATS program is automatically stopped.
(3) The operator can manually select "ATS AUTO MODE" or "OFF MODE" from TCS/DCS HMI.
(4) When the major trouble of ST or GEN. or TCS occurs, ATS program automatically changes the ATS AUTO MODE to the OFF MODE.
(5) The operator can manually select each step (stated in "2. SEQUENCE PROGRESS") after selecting "ATS AUTO MODE" PB.

2. SEQUENCE PROGRESS

The sequence is categorized to Start-UP (1) and Shut-Down (2).

(1) Start-up sequence is separated as following steps.

STEP 0 : HP TURBINE WARMING
STEP 1 : TURBINE PREPARATION
STEP 2 : TURBINE RUB-CHECK (To perform the confirmation of turbine rub-check at local side)
STEP 3 : ACCELERATION
STEP 4 : EXCITATION
STEP 5 : SYNCH. & INITIAL LOAD
STEP 6 : LOAD-UP
STEP 7 : NORMAL OPERATION

(2) Turbine shut-down sequence is only 1 step.

The basic operation of ATS is indicated on the following (1) ~ (3)

(1) When the operator selects the present or upper level step from TCS/DCS HMI, the step control program starts to execute the present step only if the STEP-IN condition (Criteria A) of the present step is satisfied.

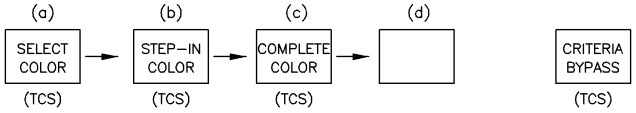
(2) During the present step execution, the step control program keeps monitoring the STEP-MONITORING condition (Criteria B) of the present step.
When an abnormal situation causes the STEP-MONITORING condition unsatisfied during the present step execution, the step control program correspondingly suspends the step execution.

However the following states are kept executing without suspending.
1) Turbine speed at the critical speed band during "ACCELERATION" step.
2) ICV to CV transfer in progress during "LOAD UP" step.

(3) When the executed result of the present step satisfies the STEP-OUT condition (Criteria D), the step control program correspondingly finishes the present step execution and then is ready to execute the next step.

As above "(1)" stated, the operator can select on upper step to start-up turbine depending on the modes, which are Ambient Cold, Cold 1, Cold 2, Warm and Hot. (Ref. DWG. 4)
The operator can skip step 0~2 at Cold 2, Warm and Hot.
The operator cannot skip step 2 at Ambient Cold and Cold 1 because the operator is required to perform the turbine Rub-Check at local side.
Additionally, step 0 and 1 are required to perform only for Ambient Cold.

3. SEQUENCE INDICATOR LAMP



(a) : This step is selected.
(b) : This STEP-IN condition is satisfied and this step control program executes turbine control and monitores the turbine status.
(c) : This step control program is completed without any upper step selected.
(d) : This step control program is completed with an upper step selected.

4. CRITERIA CHECK

ATS program categorize the permission / monitor / completion conditions into 3 criteria groups (Ref. DWG. 1), which are criteria group An, Bn and Dn. The 3 groups of criteria corresponds to each control step.

(1) Criteria group An is a group of permissive conditions to permit the STEP-IN.
(2) Criteria group Bn is a group of monitor conditions to monitor the whole step during the step execution.
(3) Criteria group Dn is a group of completion conditions to finish the executing step.
(4) If criteria An or Bn is unsatisfied, ATS program outputs the message of individual criteria to TCS HMI.

5. CRITERIA BYPASS

As above "4. (4) " stated, the operator can individually confirm the unsatisfied criteria from the TCS HMI. Furthermore the operator also can individually bypass the unsatisfied criteria. The operating procedure on TCS HMI is indicated as following steps.

(1) Select the unsatisfied criteria.
(2) Press the PB of "CRITERIA BYPASS".
(3) Confirm the unsatisfied criteria changed to be satisfied.

6. SPEED / LOAD HOLD

At the "ACCELERATION" step and the "LOAD UP" step, the acceleration of the turbine speed or the increase of the turbine load can be held on the present state and cannot be held under some conditions.

(1) The step control program can automatically hold the present turbine speed/load under following abnormal states of turbine.
-Vibration High
-Diff. expansion high
-Exhaust hood temp. high
-Condenser vacuum low
-Bearing oil press. low
-Thermal stress high
When the abnormal states of turbine are restored, ATS program flickers the "HOLD OFF" PB on TCS HMI. The operator can manually operate the "HOLD OFF" PB to continue executing the present step.
(2) The operator can manually select the "HOLD ON" PB to hold the present turbine speed/load.
(3) As above (1) (2) stated, the turbine speed /load can be hold.
But the turbine speed/load cannot be manually/ automatically held under the following turbine states.
1) Turbine speed is in the critical speed band during the "ACCELERATION" step execution.
2) Turbine control is in the transfer region from ICV to CV during the "LOAD UP" step execution.
For "1)" states, ATS program keeps increasing turbine speed to get out of critical speed band.
For "2)" states, ATS program keeps performing the ICV-to-CV transfer.

TYPE –					PAGE 3	
SERVICE DESCRIPTION OF ATS PROGRAM						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1 : 1		DWG. No. 310SC90-775	Rev. 0	Sheet 4	Date of issue 2015-05-29
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION							SEQUENTIAL FLOW	SUB LOOP OR AUX. EQUIPMENT	REMARKS
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.						
				1									
				2									
				3				1			THE CONDITION EXISTS		
				4				0			THE CONDITION DOESN'T EXIST		
				5	PERMISSIVE CONDITION								
				6									
				7									
				8									
				9									
				10									
				11									
				12									
				13									
				14									
				15									

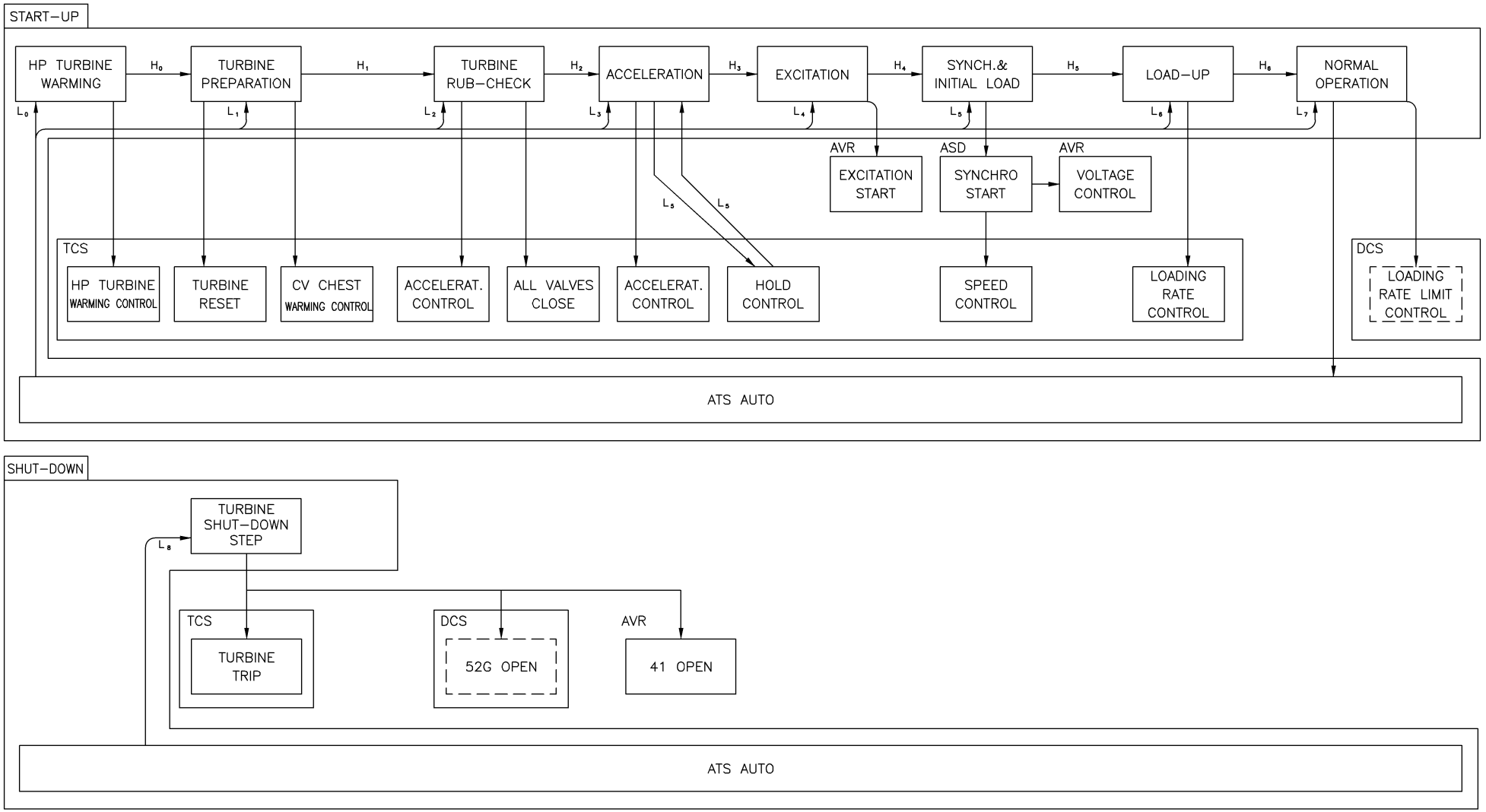
NOTE

REFER TO CONTROL STEP TRANSITION
STATE TRANSITION(SH-6)

TYPE	–				PAGE	4
SERVICE	EXPLANATION FORMAT					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-775	0	5	2015-05-29	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

CONTROL STEP TRANSITION

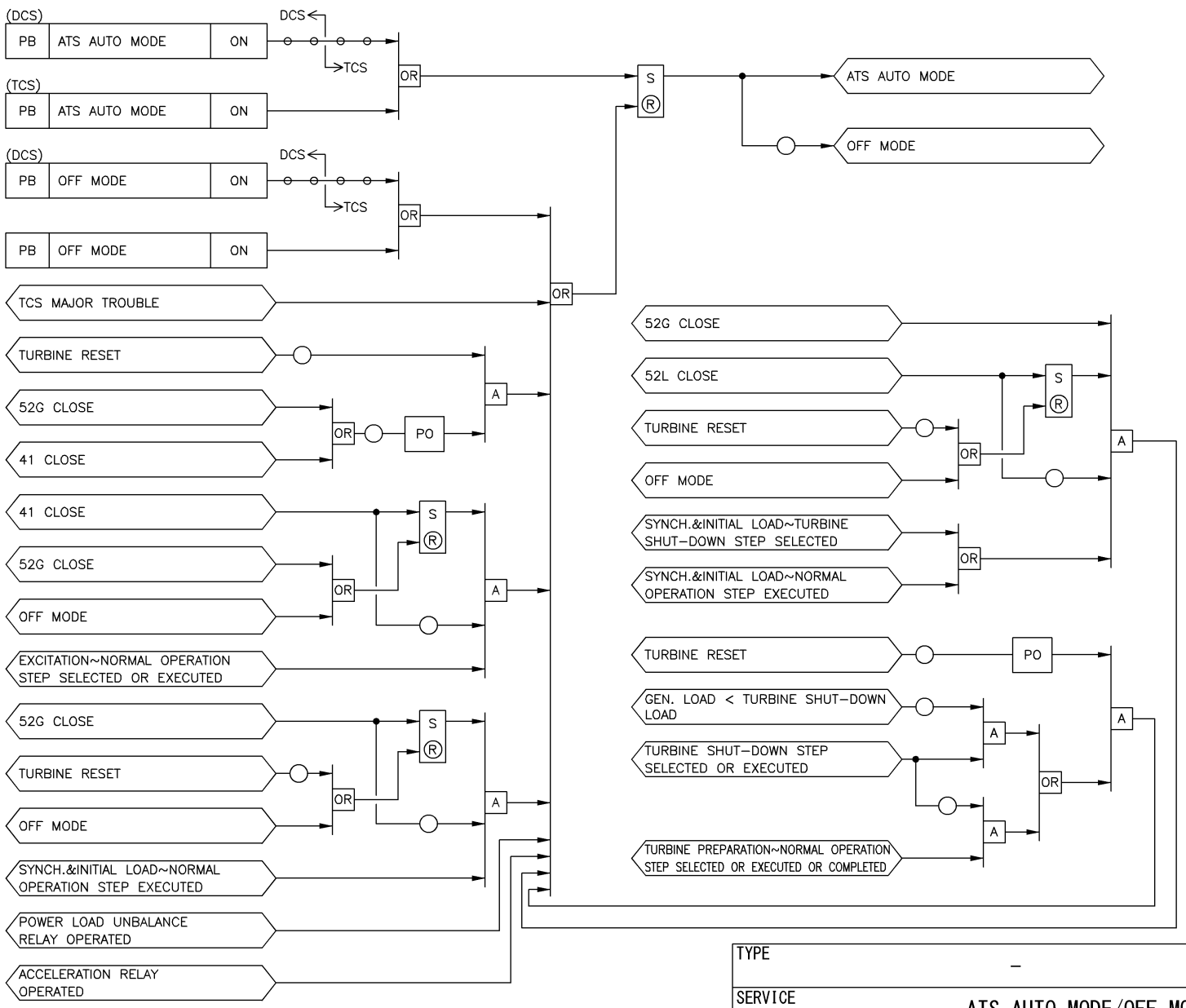


Ln : ATS MODE SIGNAL
Hn : STEP n-1 SIGNAL

TYPE	–				PAGE	5
SERVICE	CONTROL STEP TRANSITION & STATE TRANSITION					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-775	0	6	2015-05-29

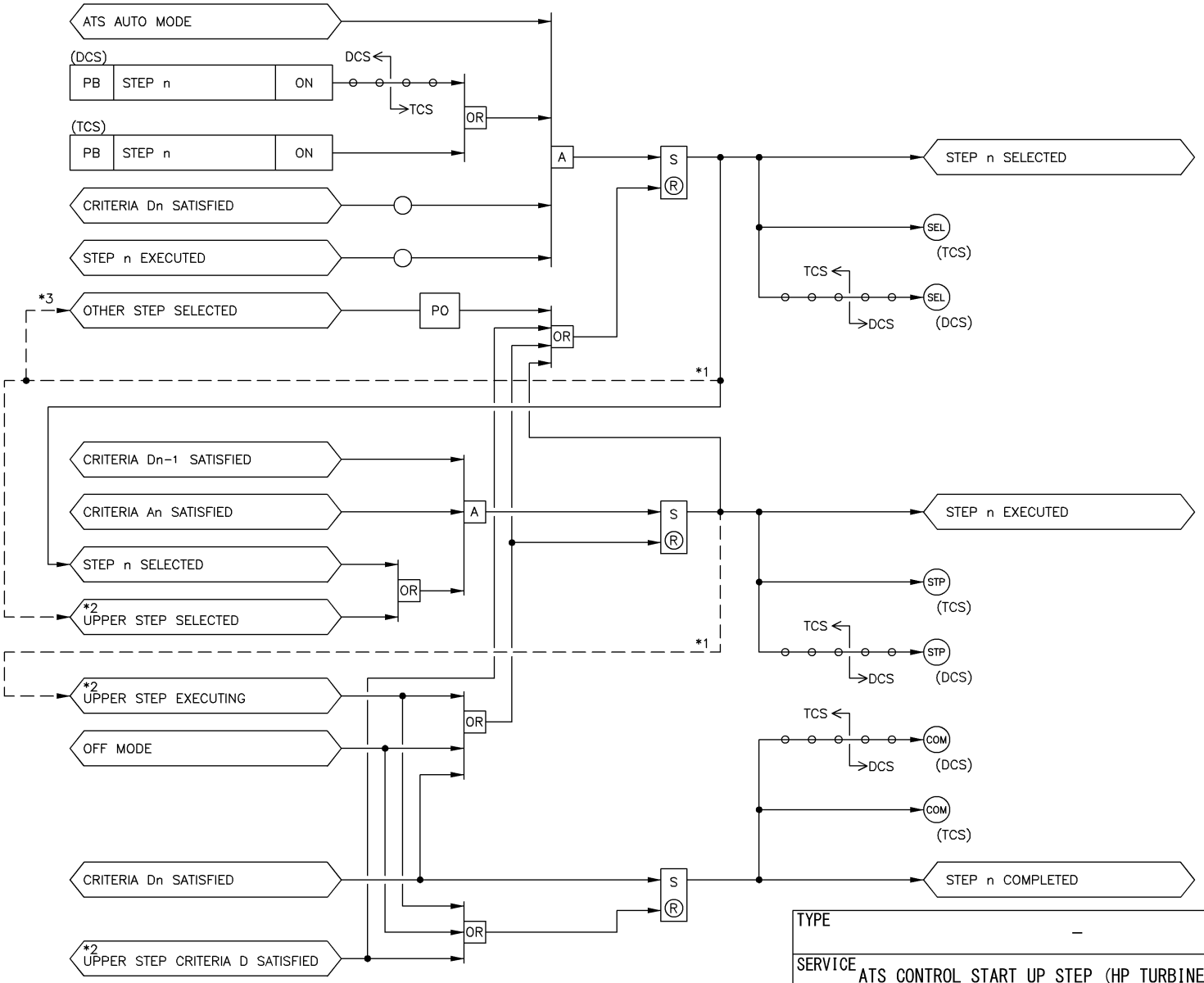
MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775



TYPE	–					PAGE	6
SERVICE	ATS AUTO MODE/OFF MODE LOGIC						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-775	0	7	2015-05-29	

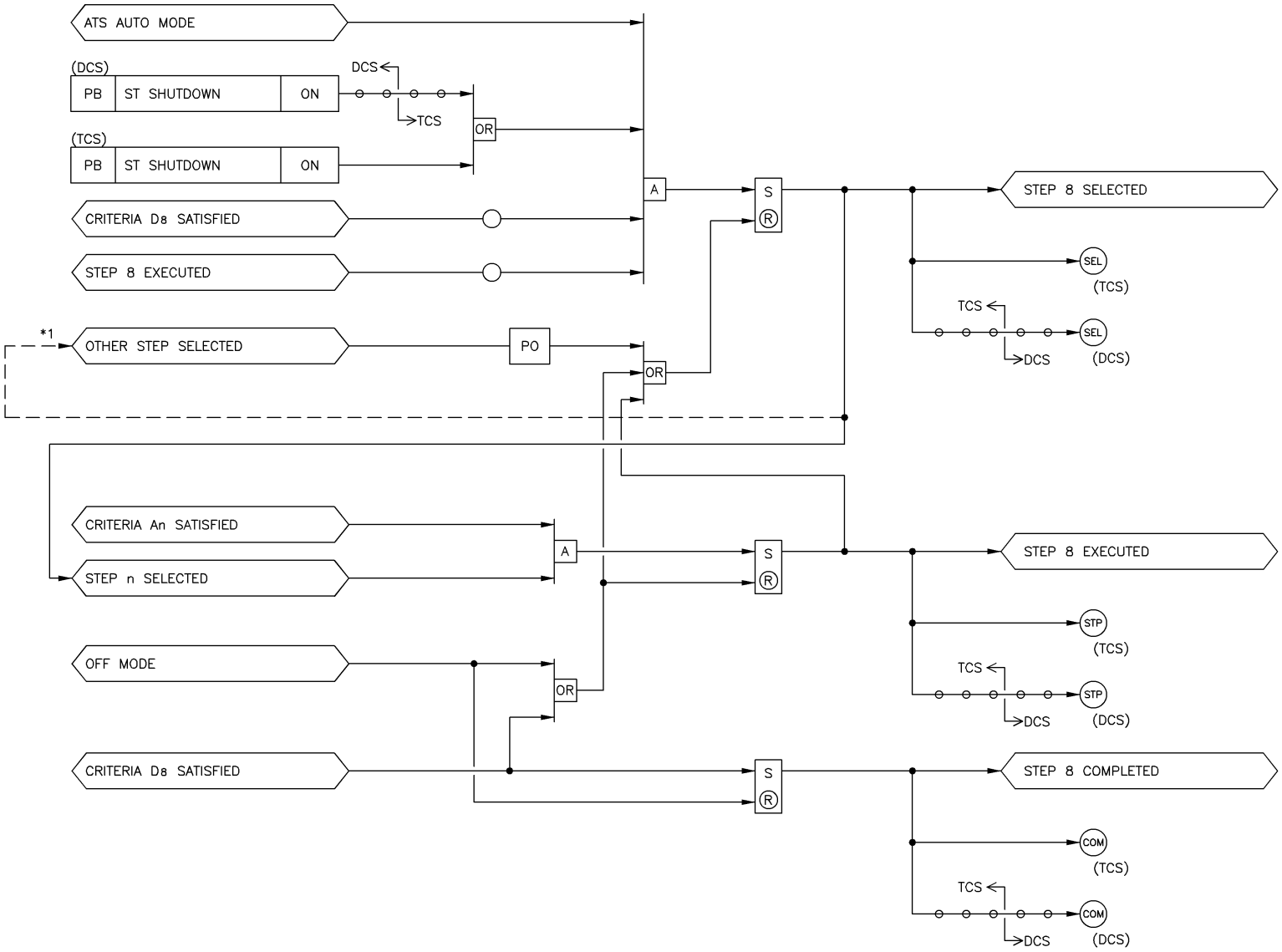
MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division



NOTE
*1. If "STEP n" is equal to "STEP 7 (NORMAL OPERATION)", the signal "----" does not exist.
*2. "UPPER STEP" means that an upper level step than the present step.
*3. "----" indicates the collective signals for "OTHER STEP".

TYPE –					PAGE 7	
SERVICE ATS CONTROL START UP STEP (HP TURBINE WARMING-NORMAL) TRANSITION LOGIC						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1		310SC90-775	0	8	2015-05-29

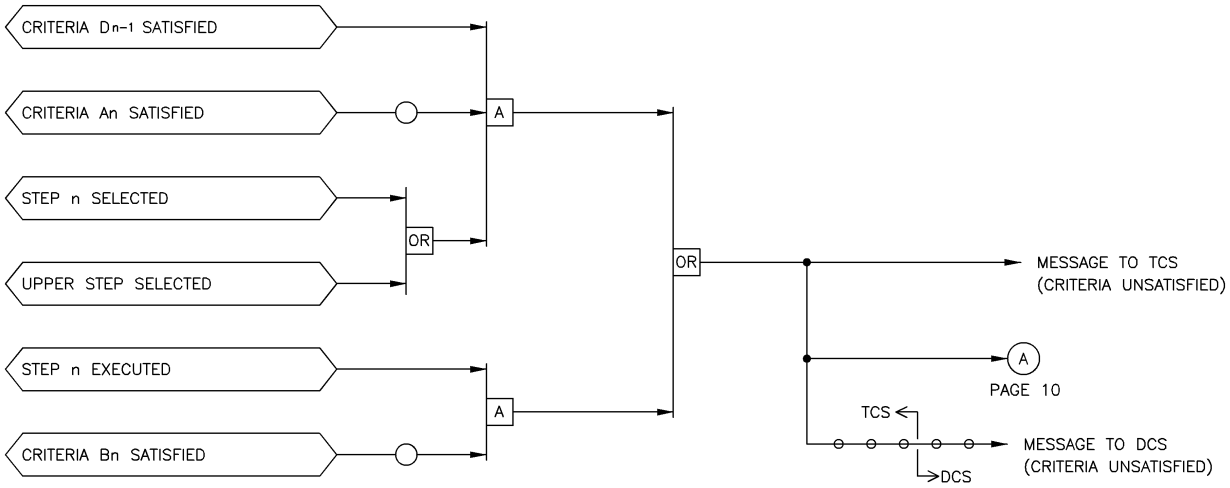
MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division



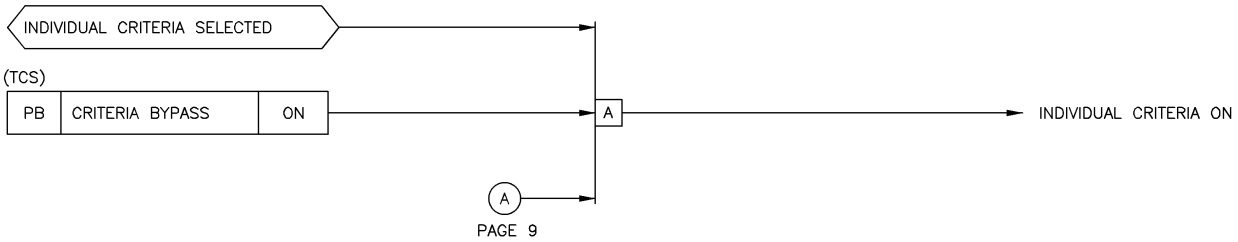
NOTE
*1. "----" indicates the collective signals for "OTHER STEP".

TYPE	–					PAGE	8
SERVICE	ATS CONTROL SHUTDOWN STEP (TURBINE SHUTDOWN) TRANSITION LOGIC						
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-775	0	9	2015-05-29	

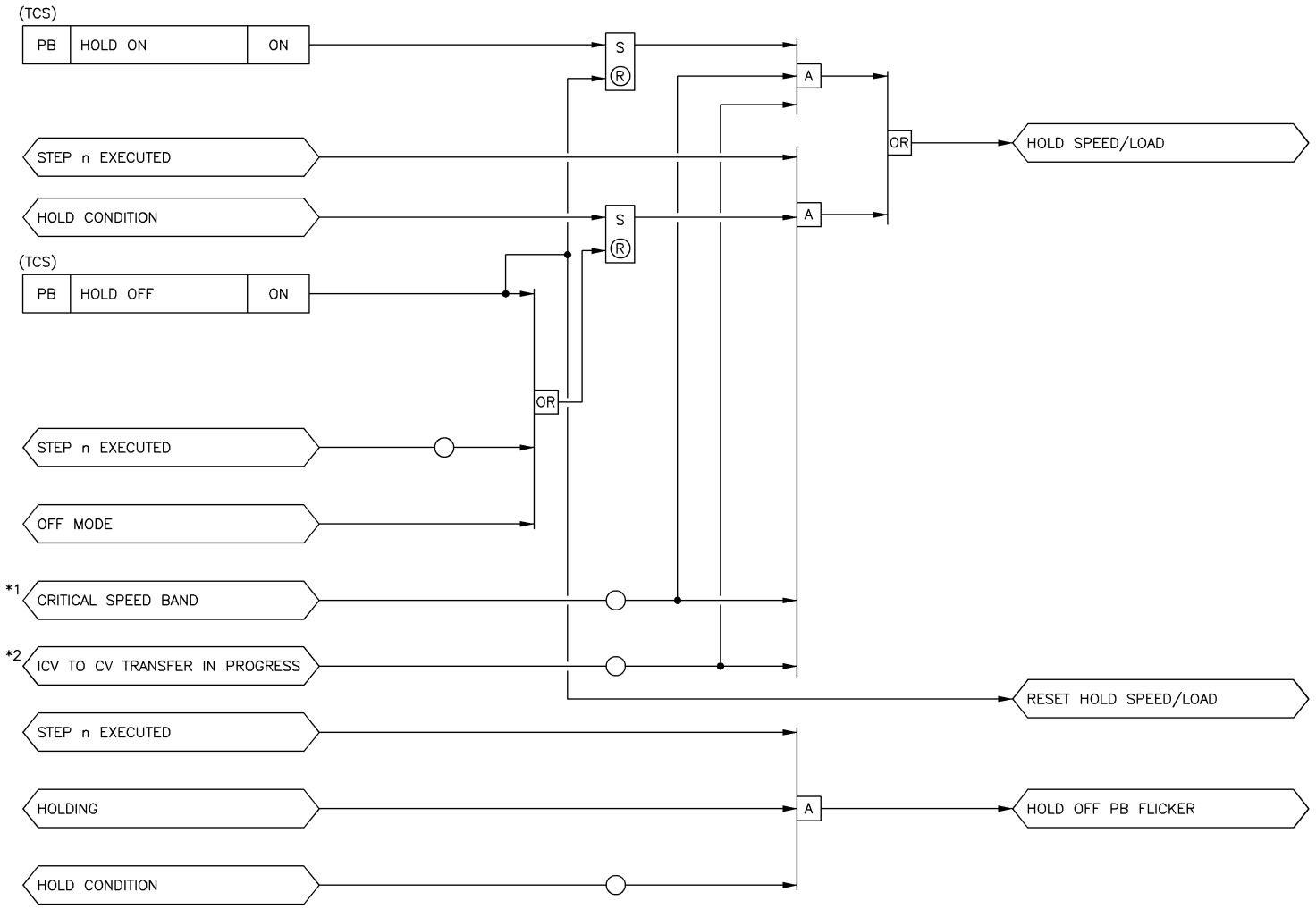
MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division



TYPE –					PAGE 9	
SERVICE CRITERIA CHECK LOGIC						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1:1		DWG. No. 310SC90-775	Rev. 0	Sheet 10	Date of issue 2015-05-29
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						



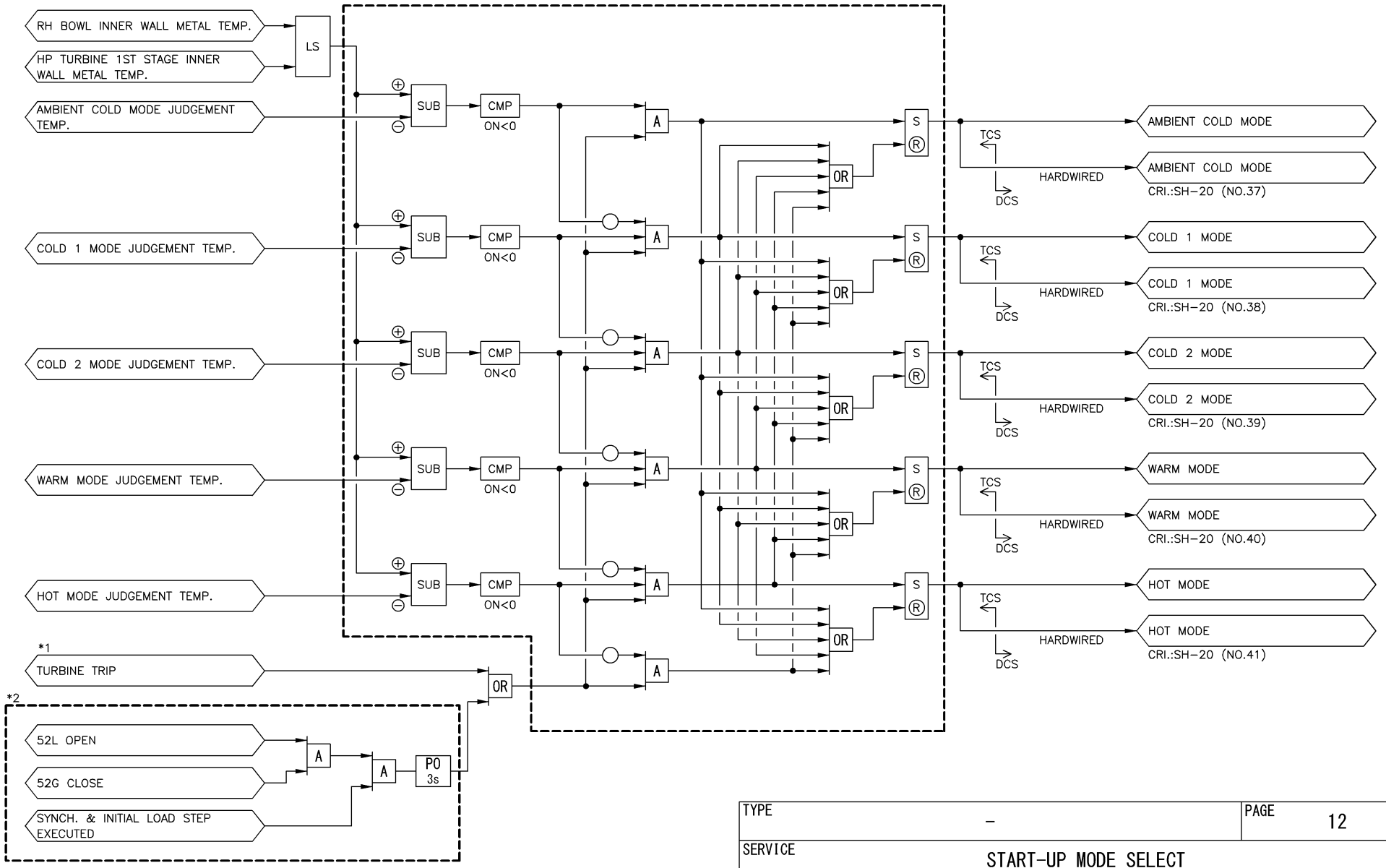
TYPE –					PAGE 10	
SERVICE CRITERIA BYPASS LOGIC						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1 : 1		DWG. No. 310SC90-775	Rev. 0	Sheet 11	Date of issue 2015-05-29
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						



NOTE
*1. At "CRITICAL SPEED BAND", "HOLD SPEED" is prohibited due to turbine protection.
*2. At "ICV to CV TRANSFER IN PROGRESS", "HOLD LOAD" is prohibited due to turbine protection.

TYPE –					PAGE 11	
SERVICE SPEED / LOAD HOLD LOGIC						
Responsible dep. Hitachi I&C Engineering Section/ Instrument & Control Department	Scale 1 : 1		DWG. No. 310SC90-775	Rev. 0	Sheet 12	Date of issue 2015-05-29
MITSUBISHI HITACHI POWER SYSTEMS, LTD. Engineering Headquarters / Plant Engineering Division						

START-UP MODE TYPICAL CALCULATING FUNCTION



NOTE

1. "*1" and "*2" are the timing conditions for start-up mode calculating function.

*1; At the normal start-up process, the start-up mode is determined at "TURBINE RESET". However we used "TURBINE TRIP" as the timing condition.

*2; For the grid reconnection from the house load operation, the start-up mode is recalculated. The related sequence is described at the "SYNCH. & INITIAL LOAD" step (PAGE 24).

TYPE	–				PAGE	12
SERVICE	START-UP MODE SELECT					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-775	0	13	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP

ITEM	FROM	LINE NO.	TO
HP TURBINE WARMING	ATS AUTO	L_o	HP TURBINE WARMING

PROGRAM ACTUATION

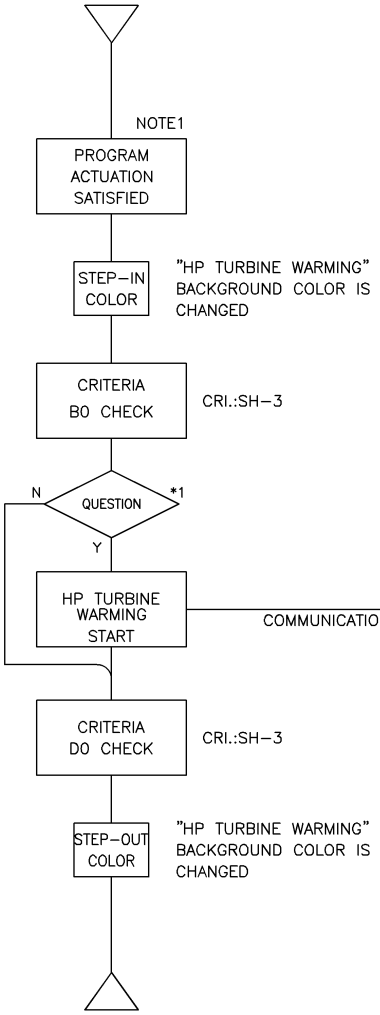
NO.	ITEM	SIGNAL STATE	LINE NO. L_o
1	TCS MAJOR TROUBLE	⌐‾	0
2	"HP TURBINE WARMING" PB ON (FROM TCS OR DCS)	⌐‾⌐	1 (or)
3	UPPER STEP SELECTED	⌐‾	1 (or)
4			
5			
6	CRITERIA AO OK	⌐‾	1
7	ATS AUTO MODE ON	⌐‾	1
8			
9			
10			
11			
12			
13			
14			
15			

NOTE

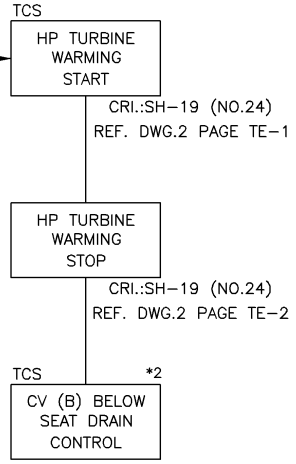
*1 QUESTION :
HP TURBINE 1ST STAGE INNER WALL METAL TEMP. ≦150°C

*2 :
After "HP TURBINE WARMING", a certain amount of drains are supposed to remain on the CV Below Seat.
This control program slightly open CB (B) to discharge the drain into the condenser and then close CV (B) to complete "HP TURBINE WARMING".

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

NOTE:

1. SEE PROGRAM ACTUATION IN THIS SHEET
2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3

TYPE	–
PAGE	13
SERVICE	HP TURBINE WARMING

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	14	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP

ITEM	FROM	LINE NO.	TO
TURBINE PREPARATION	HP TURBINE WARMING	H_0	TURBINE PREPA-RATION
	ATS AUTO	L_1	

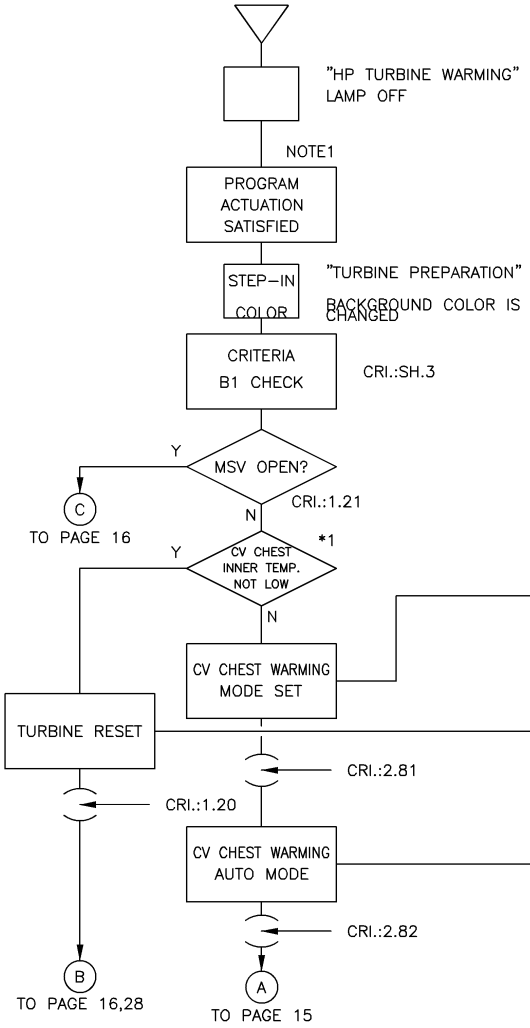
PROGRAM ACTUATION

NO.	ITEM	SIGNAL STATE	LINE NO. H_0	LINE NO. L_1
1	TCS MAJOR TROUBLE		0	0
2	"TURBINE PREPARATION" PB ON (FROM TCS OR DCS)	or	1	1
3	"RUB CHECK" SELECTED			
4				
5	ATS AUTO MODE ON		1	1
6	CRITERIA A1 OK		1	1
7	HP TURBINE WARMING FINISHED		1	
8				
9				
10				
11				
12				
13				
14				
15				

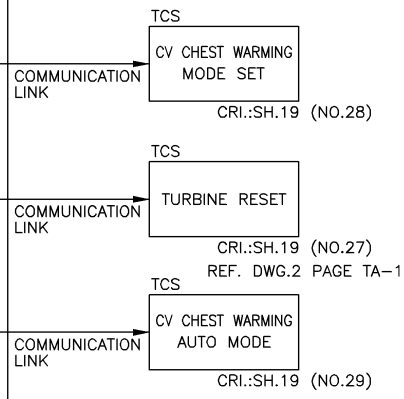
NOTE

*1 QUESTION :
CV CHEST INNER OR OUTER WALL METAL TEMP. >150°C

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

NOTE:

1. SEE PROGRAM ACTUATION IN THIS SHEET
2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3

TYPE	–
PAGE	14
SERVICE	TURBINE PREPARATION

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	15	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

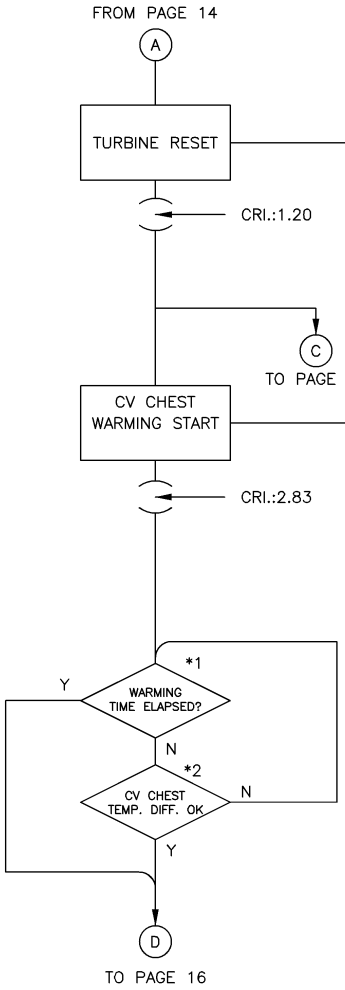
STEP				PROGRAM ACTUATION							
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.			
TURBINE PREPARATION				1							
				2							
				3							
				4							
				5							
				6							
				7							
				8							
				9							
				10							
				11							
				12							
				13							

NOTE

*1: QUESTION
ELAPSED TIME > 60MIN.

*2: QUESTION
CV CHEST WALL METAL TEMP. DIFF <50°C AND
CV CHEST OUTER WALL METAL TEMP. >180°C
(CV CHEST WALL METAL TEMP. DIFF=INNER−OUTER)

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT

TCS
TURBINE RESET
COMMUNICATION LINK
CRI.:SH.19 (NO.27)
REF. DWG.2 PAGE TA−1

TCS
CV CHEST WARMING START
COMMUNICATION LINK
CRI.:SH.19 (NO.30)

REMARKS

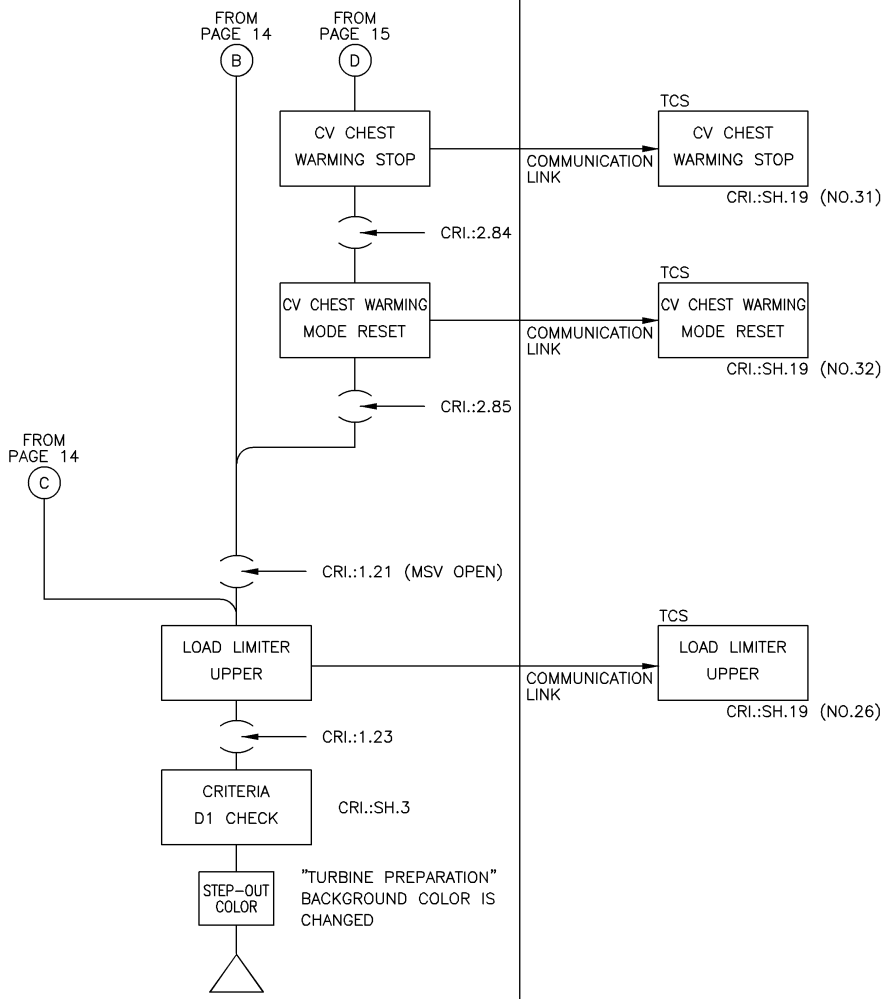
TYPE	–				PAGE	15
SERVICE	TURBINE PREPARATION					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-775	0	16	2015-05-29	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION							
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.				
TURBINE PREPARATION				1							
				2							
				3							
				4							
				5							
				6							
				7							
				8							
				9							
				10							
				11							
				12							
				13							
				14							
				15							
NOTE											

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT

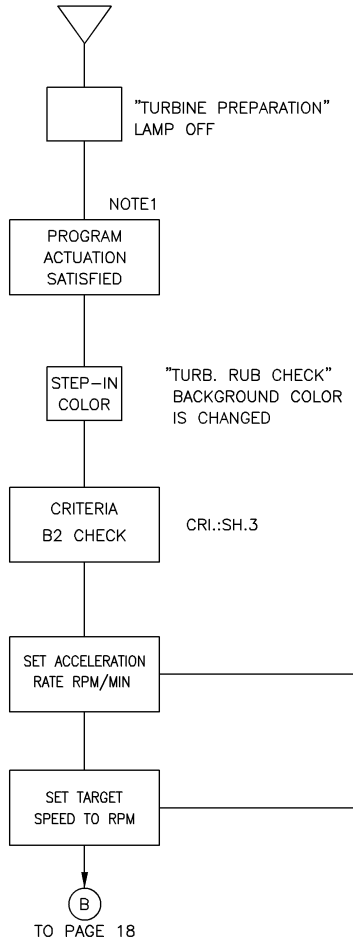
REMARKS

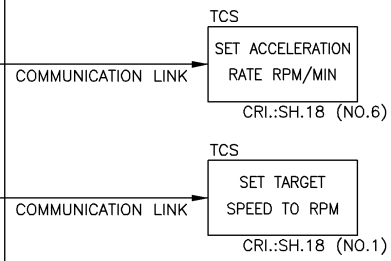
TYPE	–
PAGE	16
SERVICE	TURBINE PREPARATION
Responsible dep.	Hitachi I&C Engineering Section/ Instrument & Control Department
Scale	1 : 1
DWG. No.	310SC90-775
Rev.	0
Sheet	17
Date of issue	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION						SEQUENTIAL FLOW	SUB LOOP OR AUX. EQUIPMENT	REMARKS
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE		LINE NO. H_1	LINE NO. L_2			
TURBINE RUB CHECK	TURBINE PREPA-RATION	H_1	TURB. RUB CHECK	1	TCS MAJOR TROUBLE			0	0			NOTE: 1. SEE PROGRAM ACTUATION IN THIS SHEET 2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3
				2	ATS AUTO MODE ON			1	1			
				3								
				4	TURBINE PREPARATION FINISHED			1				
				5	UPPER STEP SELECTED		or	1	1			
				6	"RUB CHECK" PB ON (FROM TCS OR DCS)							
		L_2		7								
	ATS AUTO			8								
				9	CRITERIA A2 OK			1	1			
				10								
				11								
				12								
				13								
				14								
				15								
				NOTE								

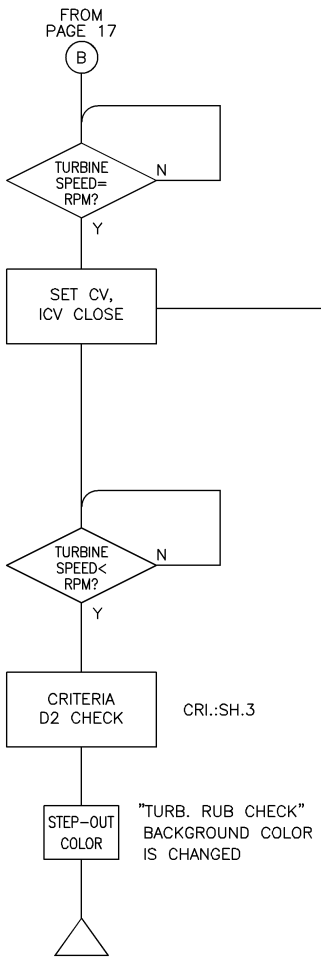
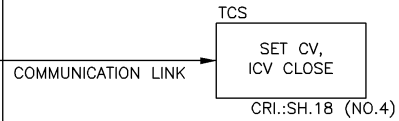




TYPE	–				PAGE	17
SERVICE	TURBINE RUB CHECK					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-775	0	18	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION							SEQUENTIAL FLOW	SUB LOOP OR AUX. EQUIPMENT	REMARKS
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.						
TURBINE RUB CHECK				1									
				2									
				3									
				4									
				5									
				6									
				7									
				8									
				9									
				10									
				11									
				12									
				13									
				14									
				15									
				NOTE									

TYPE	–				PAGE	18
SERVICE	TURBINE RUB CHECK					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	19	2015-05-29	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP

ITEM	FROM	LINE NO.	TO
ACCELERATION	TURB. RUB CHECK	H_2	ACCELE-RATION
	HOLD	I_2	
	ATS AUTO	L_3	

PROGRAM ACTUATION

NO.	ITEM	SIGNAL STATE	LINE NO. H_2	L_3	I_2
1	TCS MAJOR TROUBLE	⌐	0	0	0
2	"ACCELERATION" PB ON (FROM TCS OR DCS)	⌐⌐	or 1	1	1
3	UPPER STEP SELECTED	⌐			
4					
5					
6	HOLD CONDITION CANCELED	⌐			1
7	HOLD CONTROL ITEM 1~6 *1	⌐			1
8	TURBINE RUB CHECK FINISHED		1		
9					
10	CRITERIA A3 OK	⌐	1	1	1
11	ATS AUTO MODE ON	⌐	1	1	1
12					
13	TURBINE SPEED L.T. 3550 RPM	⌐	1	1	1
14	TURBINE TURNING	⌐	0	0	0
15					

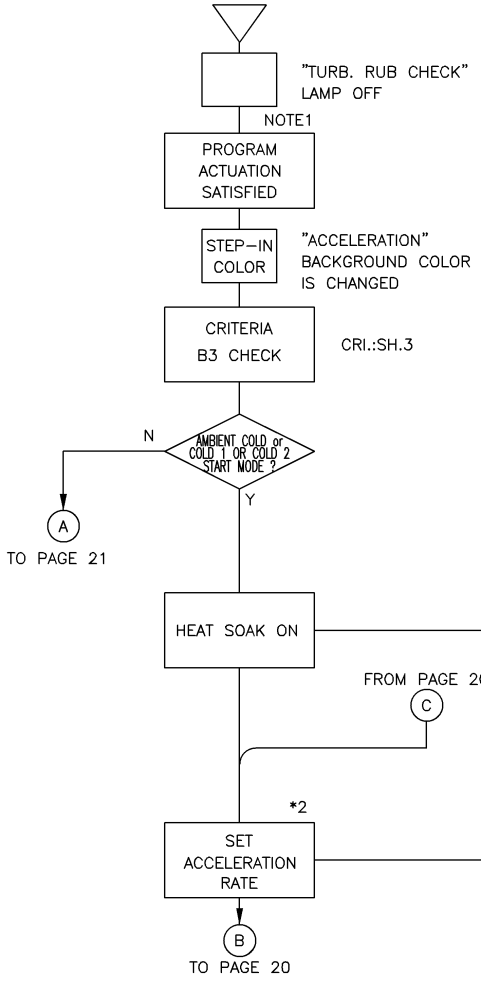
NOTE

*1: 1. VIBRATION HIGH
2. DIFF EXPANSION HIGH
3. EXHAUST HOOD TEMP. HIGH
4. CONDENSER VACUUM LOW
5. BEARING OIL PRESS LOW
6. THERMAL STRESS HIGH

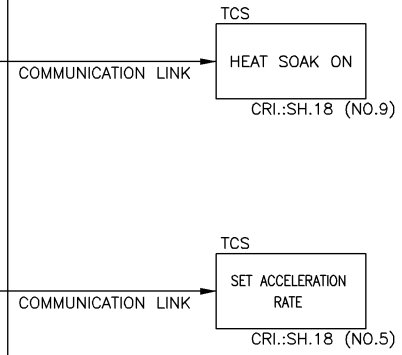
*2: MAXIMUM SELECTABLE ACCELERATION RATE DEPENDS ON START UP MODE AS FOLLOWS:

AMBIENT MODE	:	120 RPM/MIN
COLD 1 MODE	:	120 RPM/MIN
COLD 2 MODE	:	120 RPM/MIN
WARM MODE	:	180 RPM/MIN
HOT MODE	:	360 RPM/MIN

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

NOTE:

1. SEE PROGRAM ACTUATION IN THIS SHEET
2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3

TYPE	–
PAGE	19
SERVICE	ACCELERATION

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-775	0	20	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION							SEQUENTIAL FLOW	SUB LOOP OR AUX. EQUIPMENT	REMARKS
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.						
ACCELERATION				1									
				2									
				3									
				4									
				5									
				6									
				7									
				8									
				9									
				10									

NOTE

*1:

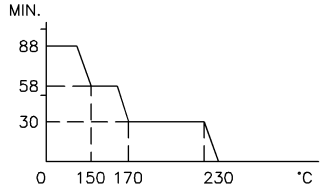


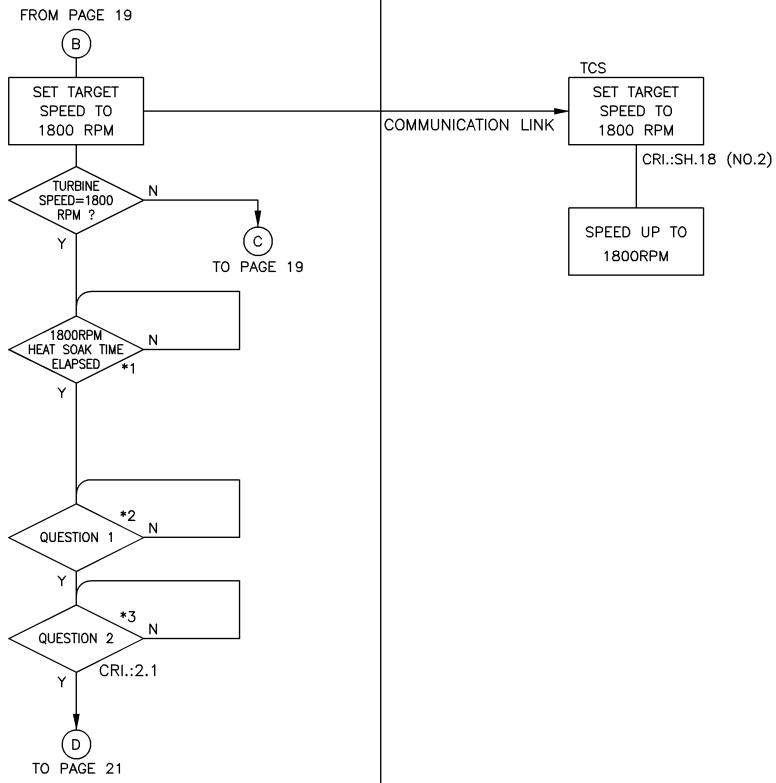
FIGURE 1. 1800rpm HEAT SOAK TIMER

One of lower value in A and B is selected at the first steam-in of turbine.

A = RH BOWL INNER WALL METAL TEMP.
B = HP TURBINE 1ST STAGE INNER WALL METAL TEMP.

*2: QUESTION 1
HP TURBINE 1ST STAGE INNER WALL METAL TEMP. >170°C

*3: QUESTION 2
TURBINE SHELL EXPANSION >3mm
(HP TURBINE SIDE)



TYPE	–				PAGE	20
SERVICE	ACCELERATION					
Responsible dep.	Scale		DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1		310SC90-775	0	21	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

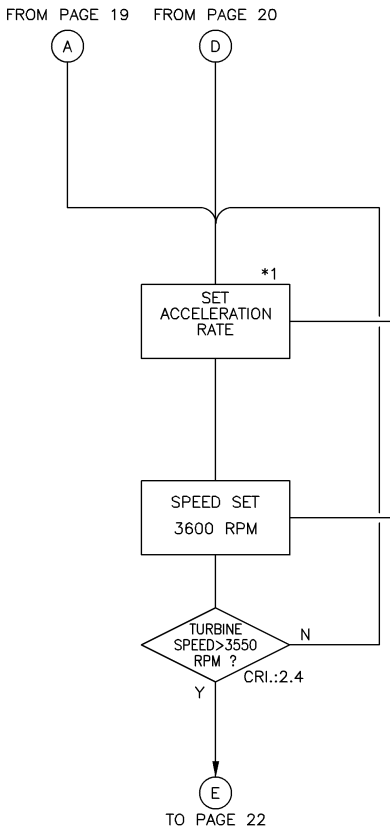
STEP				PROGRAM ACTUATION							
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.				
ACCELERATION				1							
				2							
				3							
				4							
				5							
				6							
				7							
				8							
				9							
				10							
				11							
				12							
				13							
				14							
				15							

NOTE

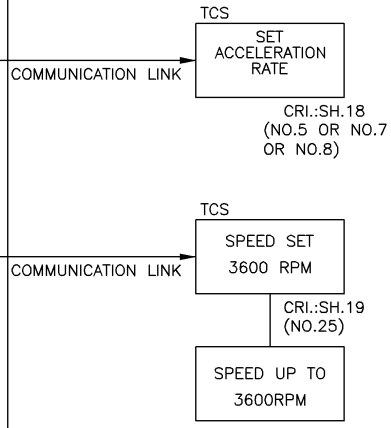
*1: MAXIMUM SELECTABLE ACCELERATION RATE DEPENDS ON START UP MODE AS FOLLOWS:

AMBIENT COLD MODE	:	120 RPM/MIN
COLD 1 MODE	:	120 RPM/MIN
COLD 2 MODE	:	120 RPM/MIN
WARM MODE	:	180 RPM/MIN
HOT MODE	:	360 RPM/MIN

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

TYPE	–
PAGE	21
SERVICE	ACCELERATION

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	22	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION							
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.			
ACCELERATION				1							
				2							
				3							
				4							
				5							
				6							
				7							
				8							
				9							
				10							

NOTE

*1:

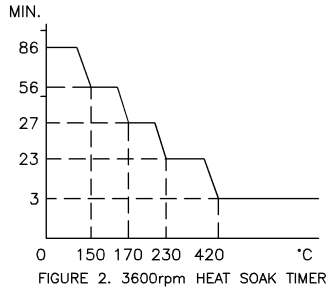
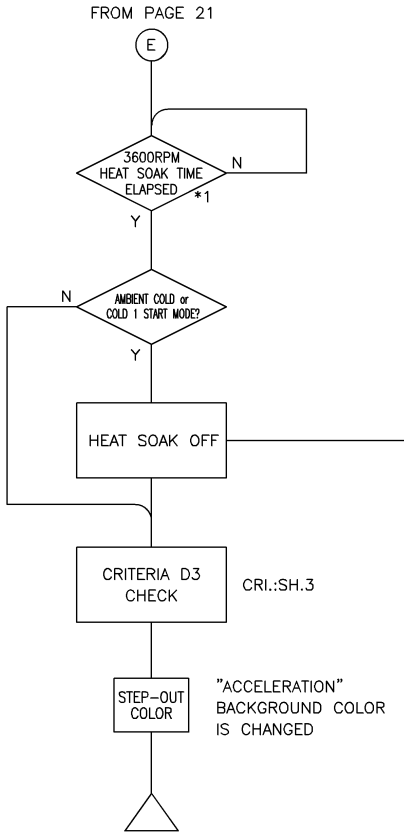


FIGURE 2. 3600rpm HEAT SOAK TIMER

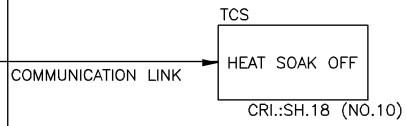
One of lower value in A and B is selected at the first steam-in of turbine.

A = RH BOWL INNER WALL METAL TEMP.
B = HP TURBINE 1ST STAGE INNER WALL METAL TEMP.

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

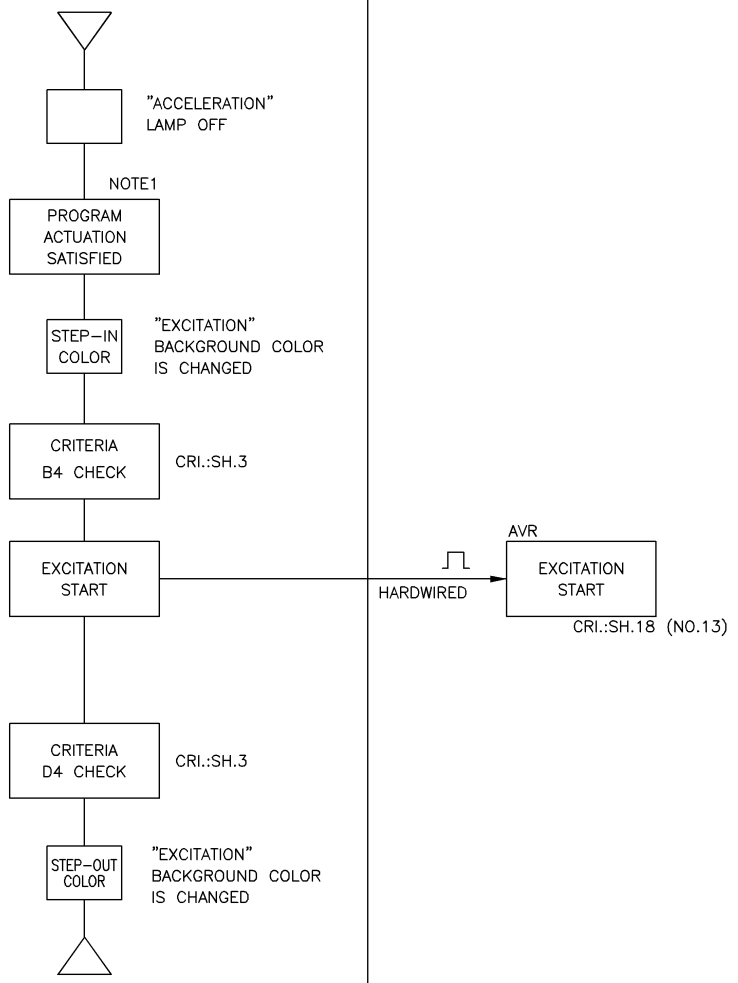
TYPE	–
PAGE	22
SERVICE	ACCELERATION

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	23	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION						SEQUENTIAL FLOW	SUB LOOP OR AUX. EQUIPMENT	REMARKS
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE		LINE NO. H_3	LINE NO. L_4			
EXCITATION	ACCELERATION	H_3	EXCITATION	1	TCS MAJOR TROUBLE			0	0			NOTE: 1. SEE PROGRAM ACTUATION IN THIS SHEET 2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3
				2	"EXCITATION" PB ON (FROM TCS OR DCS)		or	1	1			
				3	UPPER STEP SELECTED							
				4								
				5								
				6	CRITERIA A4 OK			1	1			
	ATS AUTO	L_4		7	ATS AUTO MODE ON			1	1			
				8	ACCELERATION FINISHED			1				
				9	TURBINE SPEED M.T. 3550RPM			1	1			
				10								
				11								
				12								
				13								
				14								
				15								
				NOTE								



TYPE	–				PAGE	23
SERVICE	EXCITATION					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-775	0	24	2015-05-29	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP

ITEM	FROM	LINE NO.	TO
SYNCH & INITIAL LOAD	EXCITATION	H_4	SYNCH. & INITIAL LOAD
	ATS AUTO	L_5	

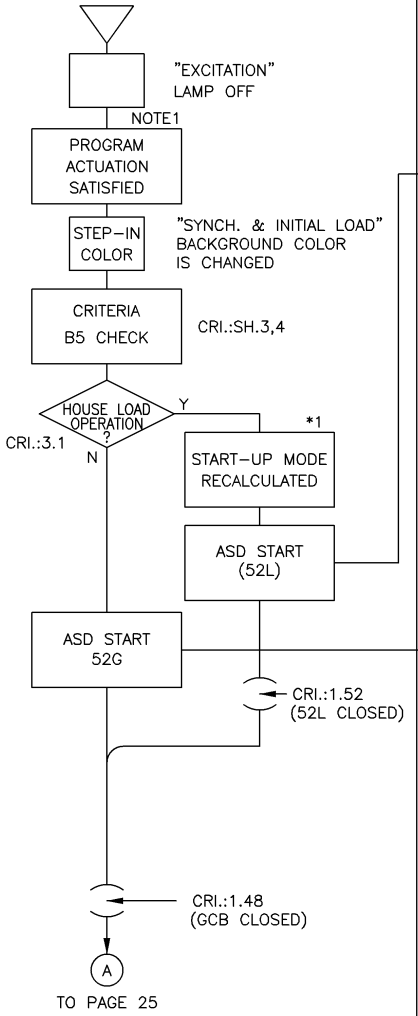
PROGRAM ACTUATION

NO.	ITEM	SIGNAL STATE	LINE NO. H_4	LINE NO. L_5
1	TCS MAJOR TROUBLE	⌐	0	0
2	"SYNCH.& INITIAL LOAD" PB ON (FROM TCS OR DCS)	⌐⌐ (or)	1	1
3	UPPER STEP SELECTED	⌐ (or)		
4				
5				
6	CRITERIA A5 OK	⌐	1	1
7	ATS AUTO MODE ON	⌐	1	1
8	EXCITATION FINISHED	⌐	1	
9				
10				
11				
12				
13				
14				
15				

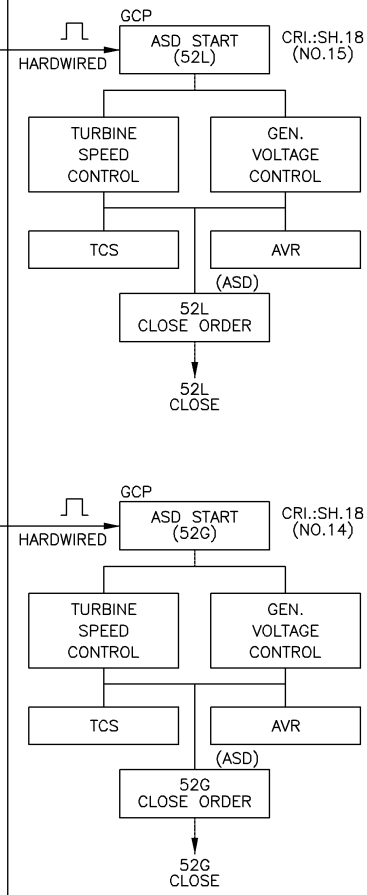
NOTE

*1: For the grid reconnection from the house load operation, the start-up mode is recalculated. For the detail, please refer to the "START-UP MODE SELECT".

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

NOTE:

1. SEE PROGRAM ACTUATION IN THIS SHEET
2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3

TYPE	–
PAGE	24
SERVICE	SYNCH. & INITIAL LOAD

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	25	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

STEP				PROGRAM ACTUATION							
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.				
SYNCH & INITIAL LOAD				1							
				2							
				3							
				4							
				5							
				6							
				7							
				8							
				9							
				10							

NOTE

*1:

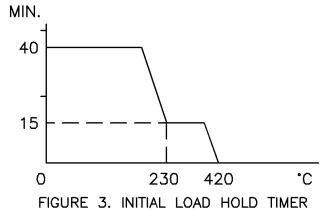


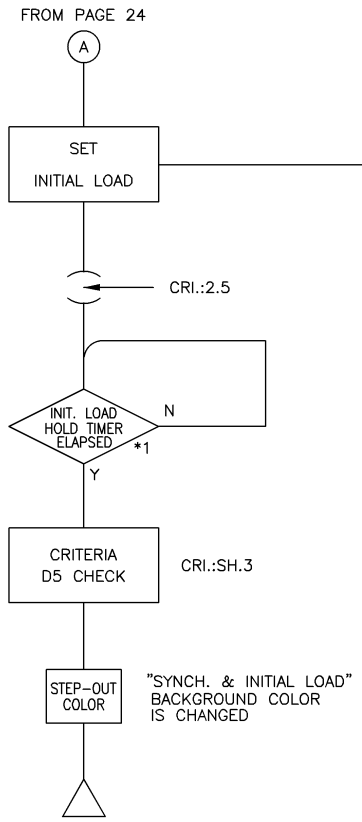
FIGURE 3. INITIAL LOAD HOLD TIMER

One of lower value in A and B is selected at the first steam-in of turbine.

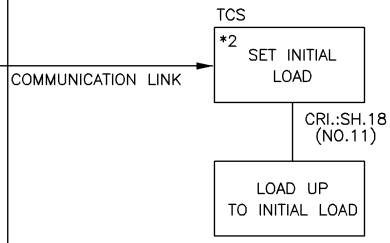
A = RH BOWL INNER WALL METAL TEMP.
B = HP TURBINE 1ST STAGE INNER WALL METAL TEMP.

*2: SET INITIAL LOAD
AMBIENT COLD : 3%
COLD 1 : 3%
COLD 2 : 3%
WARM : 3%
HOT : 5%

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

TYPE	–
PAGE	25
SERVICE	SYNCH. & INITIAL LOAD

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	26	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION						
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE		LINE NO. H_5	LINE NO. L_6	
LOAD UP	SYNCH. & INITIAL LOAD	H_5	LOAD UP	1	TCS MAJOR TROUBLE			0	0	
				2	"LOAD UP" PB ON (FROM TCS OR DCS)		or	1	1	
				3	UPPER STEP SELECTED					
				4						
				5						
				6	ATS AUTO MODE ON			1	1	
		L_6		7	CRITERIA A6 OK			1	1	
	ATS AUTO			8	SYNCH.& INITIAL LOAD FINISHED			1		
				9						
				10						
				11						
				12						
				13						
				14						
				15						

NOTE

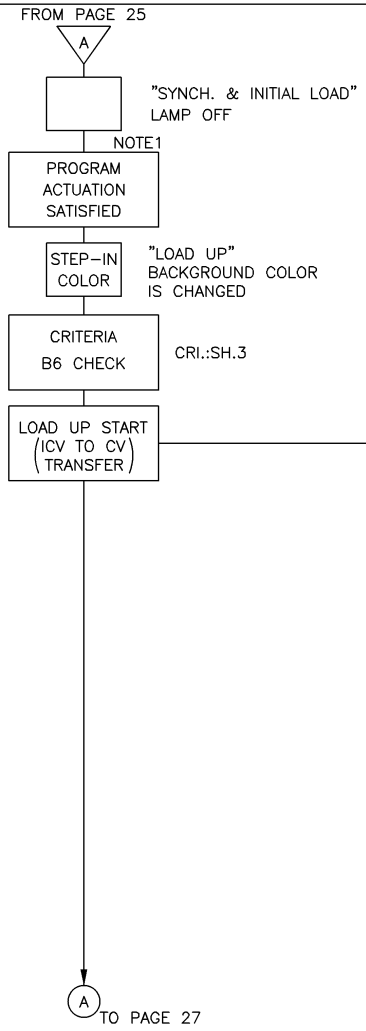
*1: 1. VIBRATION HIGH
2. DIFF EXPANSION HIGH
3. EXHAUST HOOD TEMP. HIGH
4. CONDENSER VACUUM LOW
5. BEARING OIL PRESS LOW
6. THERMAL STRESS HIGH
("THERMAL STRESS HIGH" CONDITION ARE DIFFERENT FOR START-UP AND NORMAL OPERATION CONDITION.)

MODE	THERMAL STRESS (MPa)	
	HP	IP
AMBIENT COLD	430.0	405.1
COLD 1	430.0	331.1
COLD 2	430.0	371.6
WARM	396.0	423.3
HOT	208.0	182.9
NORMAL/STOP	209.3	209.3

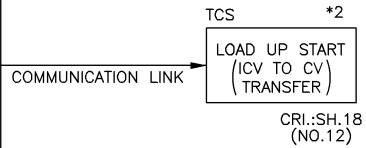
*2: LOAD UP RATE (5%/MIN) IS INCORPORATED IN TCS

*3: LOAD UP RATE (%/MIN)

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

NOTE:
1. SEE PROGRAM ACTUATION IN THIS SHEET
2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3

TYPE	–
PAGE	26
SERVICE	LOAD UP
Responsible dep.	Hitachi I&C Engineering Section/ Instrument & Control Department
Scale	1 : 1
DWG. No.	310SC90-775
Rev.	0
Sheet	27
Date of issue	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION								SEQUENTIAL FLOW	SUB LOOP OR AUX. EQUIPMENT	REMARKS
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.							
LOAD UP				1								FROM PAGE 26 A CRITERIA D6 CHECK CRI.:SH.3 STEP-OUT COLOR "LOAD UP" BACKGROUND COLOR IS CHANGED		
				2										
				3										
				4										
				5										
				6										
				7										
				8										
				9										
				10										
				11										
				12										
				13										
				14										
				15										
				NOTE										

TYPE	–				PAGE	27
SERVICE	LOAD UP					
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue	
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	28	2015-05-29	

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION					
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO. H_6	LINE NO. L_7	
NORMAL OPERATION	LOAD UP	H_6	NORM. OPER.	1	TCS MAJOR TROUBLE		0	0	
				2	"NORMAL OPERATION" PB ON (FROM TCS OR DCS)		1	1	
				3	LOAD UP FINISHED		1		
				4	ATS AUTO MODE ON		1	1	
	ATS AUTO	L_7		5					
				6	CRITERIA A7 OK		1	1	
				7					
				8					
				9					
				10					
				11					
				12					
				13					
				14					
				15					

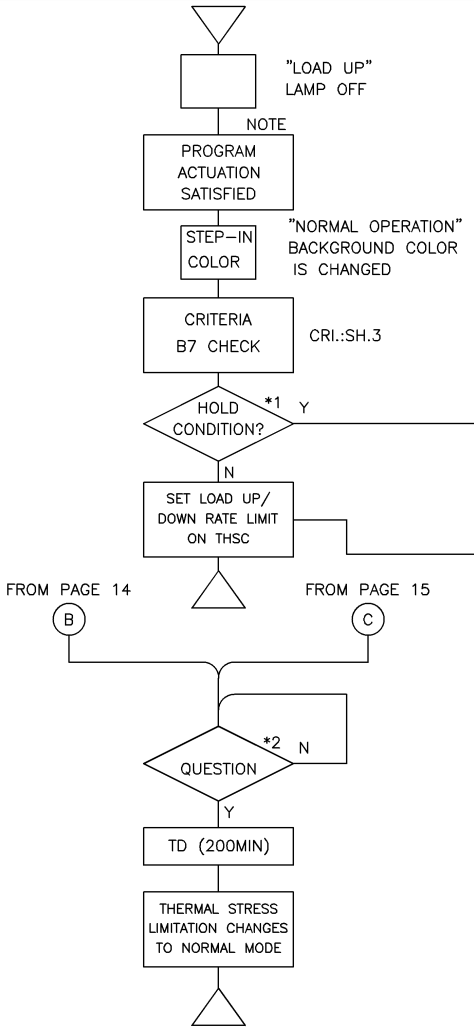
NOTE

*1: 1. VIBRATION HIGH
2. DIFF EXPANSION HIGH
3. EXHAUST HOOD TEMP. HIGH
4. CONDENSER VACUUM LOW
5. BEARING OIL PRESS LOW
6. THERMAL STRESS HIGH
("THERMAL STRESS HIGH" CONDITION ARE DIFFERENT FOR START-UP AND NORMAL OPERATION CONDITION.)

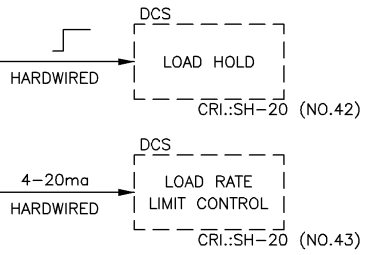
MODE	THERMAL STRESS (MPa) HP	IP
AMBIENT COLD	430.0	405.1
COLD 1	430.0	331.1
COLD 2	430.0	371.6
WARM	396.0	423.3
HOT	208.0	182.9
NORMAL/STOP	209.3	209.3

*2: HP THERMAL STRESS <209.3MPa AND IP THERMAL STRESS <209.3MPa GEN. LOAD >30%

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT



REMARKS

NOTE:
1. SEE PROGRAM ACTUATION IN THIS SHEET
2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3

TYPE	–
PAGE	28
SERVICE	NORMAL OPERATION

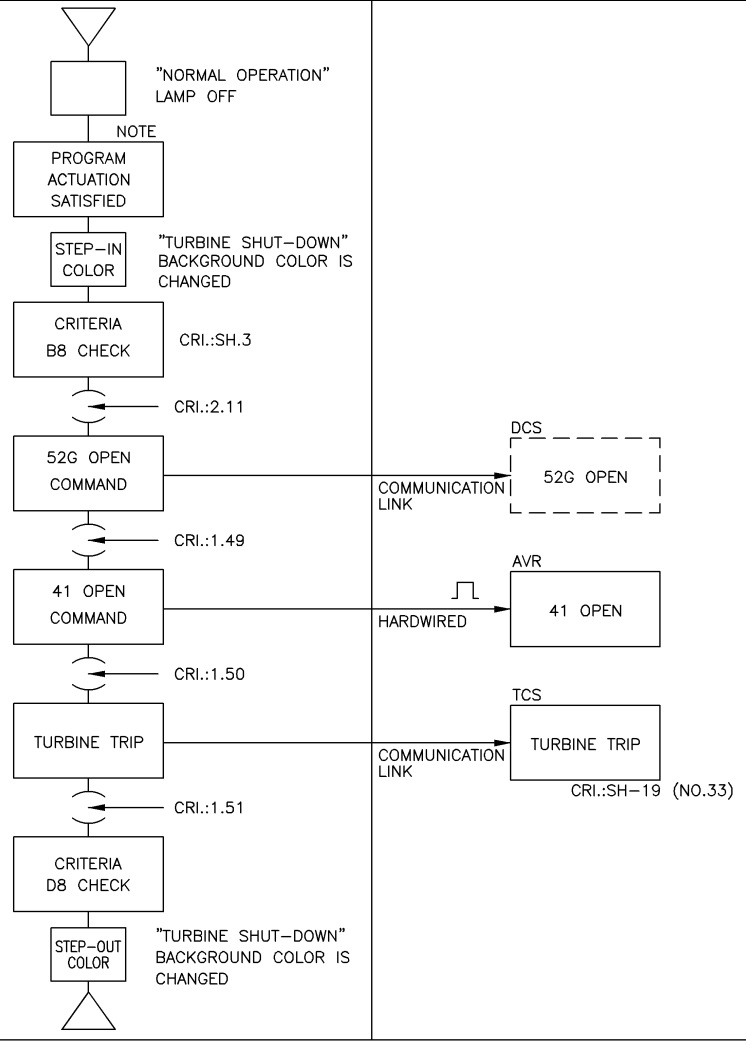
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1:1	310SC90-775	0	29	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION						
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.			
									L_8	
TURBINE SHUT-DOWN	ATS AUTO	L_8	SHUT DOWN	1	TCS MAJOR TROUBLE	⌐‾			0	
				2	"TURBINE SHUT-DOWN" PB ON (FROM TCS OR DCS)	⌐‾⌐			1	
				3						
				4	ATS AUTO MODE ON	⌐‾			1	
				5						
				6	CRITERIA A8 OK	⌐‾			1	
				7						
				8						
				9						
				10						
				11						
				12						
				13						
				14						
				15						
				NOTE						

SEQUENTIAL FLOW



SUB LOOP OR AUX. EQUIPMENT

REMARKS

NOTE:

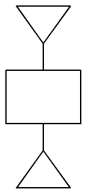
1. SEE PROGRAM ACTUATION IN THIS SHEET
2. SEE "START-UP SEQUENCE INDICATION LAMP" IN PAGE 3

TYPE	–
PAGE	29
SERVICE	TURBINE SHUT-DOWN

Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	30	2015-05-29

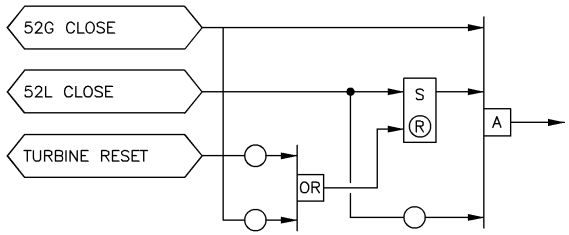
MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division

DWG. No. 310SC90-775

STEP				PROGRAM ACTUATION							SEQUENTIAL FLOW	SUB LOOP OR AUX. EQUIPMENT	REMARKS
ITEM	FROM	LINE NO.	TO	NO.	ITEM	SIGNAL STATE	LINE NO.						
HOUSE LOAD OPERATION				1	TCS MAJOR TROUBLE	⌐‾		0			 "HOUSE LOAD OPERATION" LAMP ON		
				2	ATS AUTO MODE ON	⌐‾	or	1					
				3	ATS OFF MODE ON	⌐‾							
				4	HOUSE LOAD OPERATION	⌐‾		1					
				5									
				6									
				7									
				8									
				9									
				10									
				11									
				12									
				13									
				14									
				15									

NOTE

HOUSE LOAD OPERATION INDICATION LOGIC



TYPE	–			PAGE	30
SERVICE	HOUSE LOAD OPERATION INDICATION				
Responsible dep.	Scale	DWG. No.	Rev.	Sheet	Date of issue
Hitachi I&C Engineering Section/ Instrument & Control Department	1 : 1	310SC90-775	0	31	2015-05-29

MITSUBISHI HITACHI POWER SYSTEMS, LTD.
Engineering Headquarters / Plant Engineering Division